

Kenya Financial Inclusion Risk Prediction

Project Objective

1

Predict individuals at risk of financial exclusion in Kenya.

2

Use machine learning to support financial inclusion policies.

3

Provide transparent and explainable predictions.

4

Enable targeted interventions for vulnerable groups.

Business Problem



Millions remain outside formal financial systems.



Financial exclusion limits economic growth and resilience.



Manual identification of high-risk groups is inefficient.

Dataset Overview



- Source: Kenya FinAccess 2021 Survey.
- Contains demographic, socioeconomic and behavioral indicators.
- Features include education, age, location, employment and financial behavior.
- Target variable: Financially Excluded vs Included.

Data Preparation Pipeline

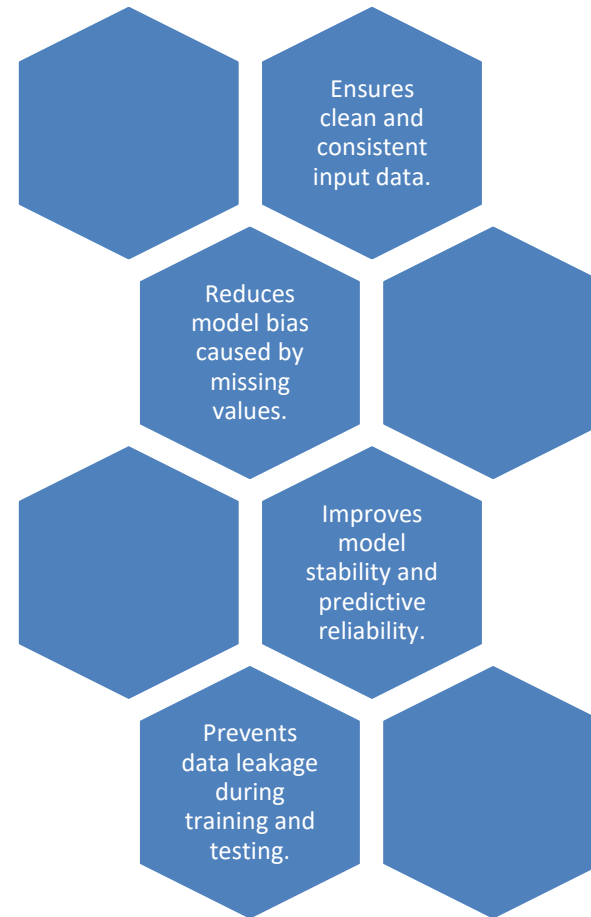
Missing categorical values filled using most frequent category.

Missing numerical values filled using median imputation.

OneHotEncoder transformed categorical variables.

StandardScaler normalized numerical features.

Why Preprocessing Matters



Models Evaluated

- Dummy Classifier: Baseline comparison model.
- Logistic Regression: Simple and interpretable classifier.
- Random Forest: Ensemble model capturing non-linear patterns.
- Gradient Boosting: Sequential boosting model optimized for performance.

Why Multiple Models Were Tested

- Different algorithms capture different data patterns.
- Linear models improve interpretability.
- Tree-based models improve complex pattern detection.
- Comparing models ensures robust final model selection.



Evaluation Metrics Explained

- Accuracy: Overall percentage of correct predictions.
- Recall: Ability to correctly identify excluded individuals.
- Precision: Reliability of positive predictions.
- F1 Score: Balance between recall and precision.
- ROC-AUC: Measures overall classification quality.

Understanding Model Trade-Offs



- High accuracy alone is not enough in imbalanced datasets.
- Recall was prioritized to avoid missing excluded individuals.
- False negatives are costly because vulnerable people may be ignored.
- Balanced metrics were used for fair model comparison.



Best Model Selection

- Gradient Boosting achieved the strongest overall predictive performance.
- It captured complex nonlinear relationships in the data.
- Provided better balance between accuracy and generalization.
- Selected as final production-ready model for explainability analysis.

Why Gradient Boosting Performed Best



Builds trees sequentially to correct previous errors.



Handles mixed feature interactions effectively.



Reduces bias while maintaining strong predictive power.



Works well with structured socioeconomic survey data.

Model Explainability with SHAP



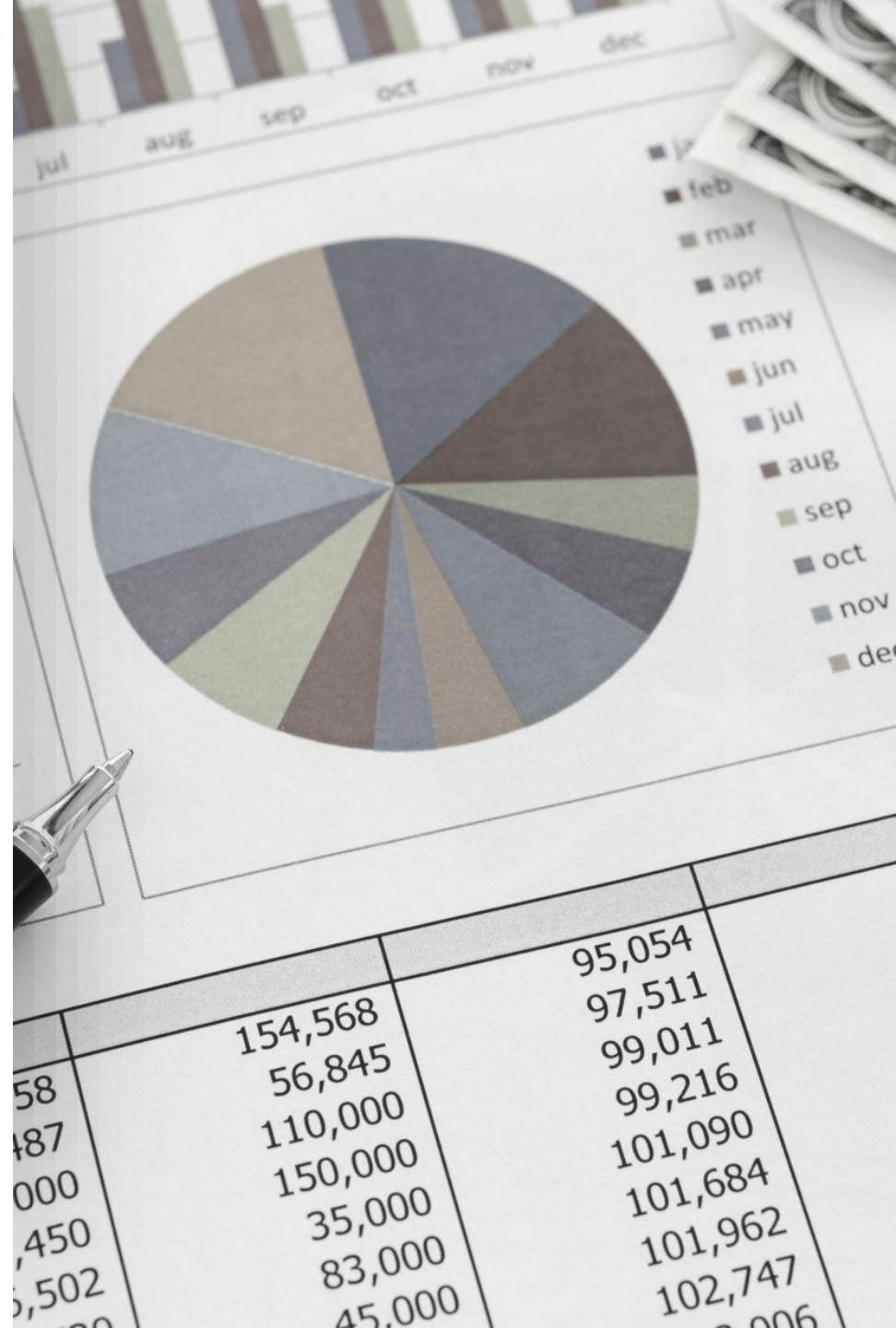
- SHAP explains how features influence predictions.
- Shows why an individual was classified as excluded.
- Improves transparency and trust .
- Supports policy interpretation and stakeholder confidence.

Key SHAP Insights

- Low education increased exclusion risk significantly.
- Rural residence strongly contributed to exclusion predictions.
- Youth populations showed higher vulnerability.
- Socio-economic status influenced financial inclusion outcomes.

Feature Importance Analysis

- Education level was among the strongest predictors.
- Employment and income indicators improved inclusion probability.
- Geographic location influenced access to financial services.
- Behavioral variables captured financial engagement patterns.

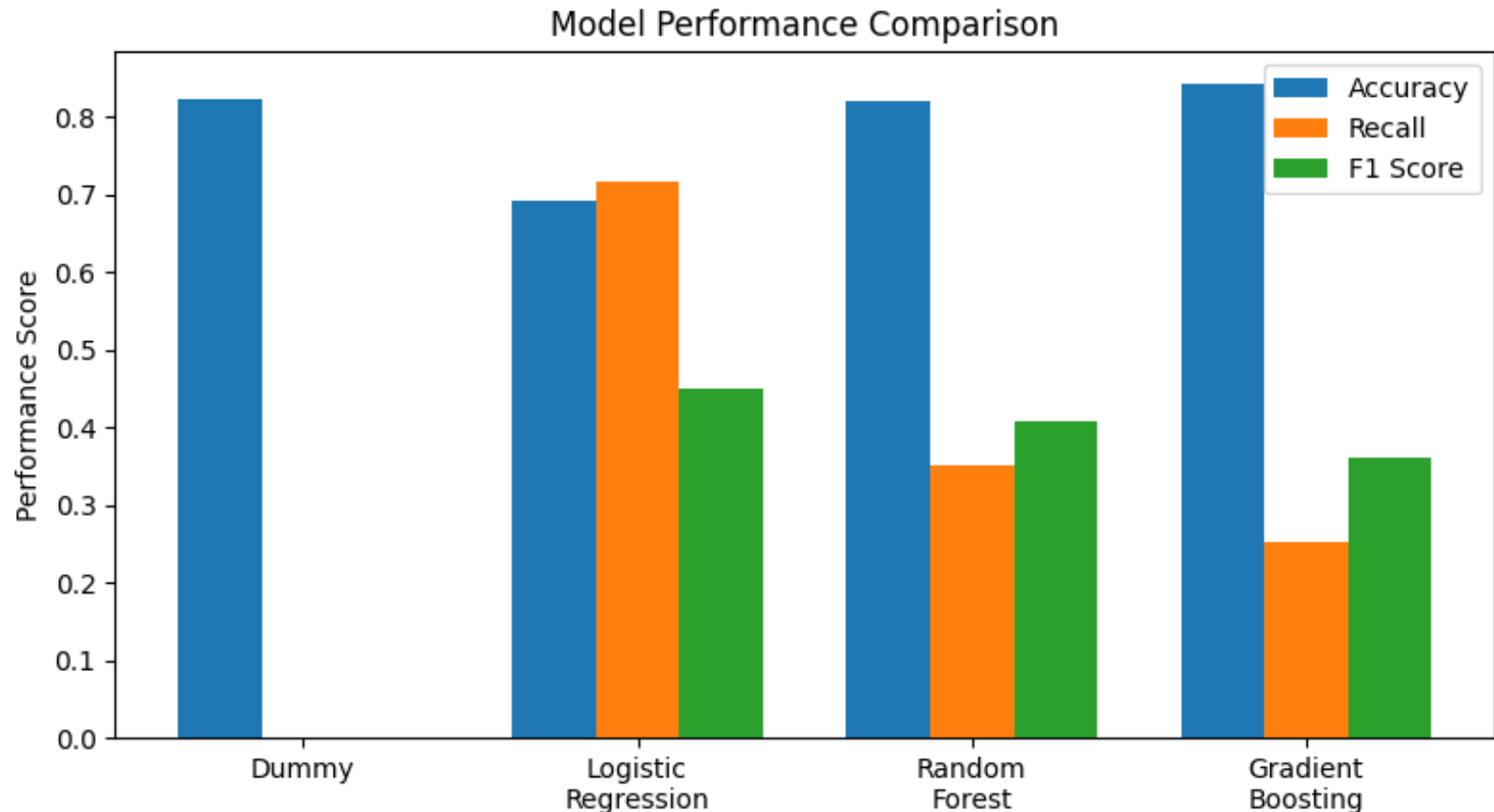


Project Impact

- Supports evidence-based financial inclusion policies.
- Can help banks target underserved populations.
- Improves transparency in predictive decision systems.
- Demonstrates responsible and explainable AI application in Kenya.



Model Performance Comparison



Conclusion

- Machine learning successfully identified financial exclusion patterns.
- Gradient Boosting delivered the strongest overall predictive performance.
- SHAP explainability improved model transparency and trust.
- The project demonstrates to improve and support financial inclusion in Kenya.

